Yuan Chen

Contact Information

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Employment

Division of Applied Mathematics, Brown University — Providence, RI
Prager Assistant Professor — 2026/7–present

Education

The Ohio State University — Columbus, Ohio
Ph.D. in Mathematics — 2021–2026
Advisor: Prof. Dongbin Xiu

The George Washington University — Washington, DC
M.S. in Statistics, GPA: 4.0/4.0 — 2019–2021

Hohai University — Nanjing, China
B.E. in Environmental Science, GPA Rank: 1^{st}/82 — 2015–2019

Research Interests

- Data-driven modeling of stochastic dynamics
- Generative models and applications in scientific computing
- Design and analysis of Finite Element Method and discontinuous Galerkin Method

Publications

Preprints

16. **Y. Chen**, W. Mao, and D. Xiu. Data-driven Effective Modeling of Stochastic Chemical Reaction Networks, (**2026**), *arXiv preprint.*

Refereed Journal Publications

15. **Y. Chen**, and X. Zhang. A High-Order Immersed C^0 Interior Penalty Method for Biharmonic Interface Problems, *Journal of Scientific Computing*, 108(**2026**), 38.

14. **Y. Chen**, and D. Xiu. Modeling Unknown Stochastic Dynamical System Subject to External Excitation, *SIAM Journal on Scientific Computing*, 48(**2026**), C216-C239.

13. Y. Liu, **Y. Chen**, D. Xiu, and G. Zhang. A Training-Free Conditional Diffusion Model for Learning Stochastic Dynamical Systems, *SIAM Journal on Scientific Computing*, 47(**2025**), C1144-C1171.

12. **Y. Chen**, D. Xiu and X. Zhang. On Enforcing Non-negativity in Polynomial Approximations in High Dimensions, *SIAM Journal on Scientific Computing*, 47(**2025**), A866-A888.

11. Z. Xu, **Y. Chen**, and D. Xiu. Chebyshev Feature Neural Network for Accurate Function Approximation, *Journal of Machine Learning for Modeling and Computing*, 5(**2025**), 1-14.

10. **Y. Chen**, and Y. Xing. Optimal Error Estimates of Ultra-weak Discontinuous Galerkin Methods with Generalized Numerical Fluxes for Multi-dimensional Convection-Diffusion and Biharmonic Equations, *Mathematics of Computation*, 93(**2024**), 2135-2183.

9. Z. Xu, **Y. Chen**, Q. Chen and D. Xiu. Modeling Unknown Stochastic Dynamical System via Autoencoder, *Journal of Machine Learning for Modeling and Computing*, 5(**2024**), 87-112.

8. **Y. Chen**, and D. Xiu. Learning Stochastic Dynamical System via Flow Map Operator, *Journal of Computational Physics*, 508(**2024**), 112984.

7. **Y. Chen**, and D. Xiu. Data-driven Effective Modeling of Multiscale Stochastic Dynamical Systems, *Applied Mathematics for Modern Challenges*, 2(**2024**), 348-364.

6. **Y. Chen**, and X. Zhang. Solving Navier-Stokes Interface Problems with Fixed/Moving Interfaces on Unfitted Meshes, *Journal of Scientific Computing*, 98(**2024**), 19.

5. V. Churchill, **Y. Chen**, Z. Xu, and D. Xiu. DNN Modeling of Partial Differential Equations with Incomplete Data, *Journal of Computational Physics*, 493(**2023**), 112502.

4. **Y. Chen**, S. Hou, and X. Zhang. Semi and Fully Discrete Analysis for An Immersed Finite Element Method for Elastodynamic Interface Problems, *Computers and Mathematics with Applications*, 147(**2023**), 92-110.

3. **Y. Chen** and X. Zhang. A $\mathcal{P}_2$-$\mathcal{P}_1$ Partially Penalized Immersed Finite Element Method for Stokes Interface Problems, *International Journal of Numerical Analysis and Modeling*, 18(**2021**), no. 1, 120-141.

2. **Y. Chen**, S. Hou, and X. Zhang. A Bilinear Partially Penalized Immersed Finite Element Method for Elliptic Interface Problems with Multi-domains and Triple Junction Points, *Results in Applied Mathematics*, 8(**2020**), 100100.

1. **Y. Chen**, S. Hou, and X. Zhang. An Immersed Finite Element Method for Elliptic Interface Problems with Multi-domain and Triple Junction Points, *Advances in Applied Mathematics and Mechanics*, 11(**2019**), no. 5, 1005-1021.

Conferences and Seminars

- 2027 AMS Spring Southeastern Sectional Meeting, Birmingham, Mar 2027
- 2027 SIAM Conference on Computational Science and Engineering, Pittsburgh, Feb 2027
- 2026 SIAM Conference on Mathematics of Data Science, Salt Lake City, Nov 2026
- 2026 SIAM New York-New Jersey-Pennsylvania Section Annual Conference, Rutgers University, Oct 2026
- ICERM Workshop on Extreme and Rare Events, Providence, Sep 2026
- 2026 SIAM Annual Meeting, Cleveland, Jul 2026
- 2026 SIAM Conference on Uncertainty Quantification, Minneapolis, Mar 2026
- 18th U. S. National Congress on Computational Mechanics, Chicago, Jul 2025
- 2025 SIAM Conference on Computational Science and Engineering, Fort Worth, Mar 2025

- 2024 SIAM Conference on Mathematics of Data Science, Atlanta, Oct 2024
- The 9th Annual Meeting of SIAM Central States Section, University of Missouri - Kansas City, Oct 2024
- 2024 SIAM Conference on Imaging Science, Atlanta, May 2024
- Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability Conference 2024, Chicago, May 2024
- 2024 SIAM Conference on Uncertainty Quantification, Trieste, Italy, Feb 2024
- The 8th Annual Meeting of SIAM Central States Section, University of Nebraska Lincoln, Oct 2023
- 17th U. S. National Congress on Computational Mechanics, Albuquerque, Jul 2023
- The 7th Annual Meeting of SIAM Central States Section, Oklahoma State University, Oct 2022
- 2022 SIAM Annual Meeting, Pittsburgh, Jul 2022
- The 6th Annual Meeting of SIAM Central States Section, University of Kansas, Oct 2021

Teaching Experiences

Ohio State University

Spring 2023	Recitation	MATH 1151 (Calculus I)
Fall 2022	Recitation	MATH 1151 (Calculus I)

George Washington University

Fall 2020	Recitation	MATH 1051 (Finite Math for the Social and Management Sciences)

Professional Services

Seminar Series Organization

- OSU Student Computational Mathematics Seminar, 2022-2024

Referee Services

- Reviewer for Applied Numerical Mathematics
- Reviewer for Communications in Computational Physics
- Reviewer for Electronic Research Archive
- Reviewer for International Journal of Numerical Analysis and Modeling
- Reviewer for Neurocomputing
- Reviewer for Journal of Computational and Applied Mathematics
- Reviewer for Journal of Computational Physics
- Reviewer for Journal of Machine Learning for Modeling and Computing
- Reviewer for Journal of Scientific Computing
- Reviewer for Numerical Methods for Partial Differential Equations

- Reviewer for Physica D: Nonlinear Phenomena
- Reviewer for Results in Applied Mathematics
- Reviewer for SIAM Journal on Scientific Computing

Scholarships & Certificates

- OSU Dissertation Year Fellowship 2025
- SIAM Travel Award 2022-2024
- OSU Distinguished University Fellowship 2021
- GWU Award of Graduate Assistantship 2020

Skills

MATLAB, Python (PyTorch, TensorFlow)